BOUNDED ABOVE COMPLEXES RESOLVE BY TERMWISE SURJECTIVE QUASI-ISOMORPHISMS FROM P

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5535-5601, following the statement at 5518-5533.

Editorial note. The induction-step verification is 'The reader easily checks' (5583): mechanical check of the fully displayed IH data, recorded.

Allowed standard prerequisites (editorial).

- Truncation quasi-isomorphism. homology-section-truncations, cross-chapter reference at proof line 5597 (tags mechanism). Supplies tau_{<= a}K -> K as a quasi-isomorphism in part (2).

Source reference. Original reference 0118: focused-seeds/geometry/S9-der-subcategory-left-resolution/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{P} \subset \mathrm{Ob}(\mathcal{A})$ be a subset containing 0 such that every object of $\mathcal{A}$ is a quotient of an element of $\mathcal{P}$. Let $a \in \mathbf{Z}$.*

(1) *Given $K^\bullet$ with $K^n = 0$ for $n > a$ there exists a quasi-isomorphism $P^\bullet \to K^\bullet$ with $P^n \in \mathcal{P}$ and $P^n \to K^n$ surjective for all n and $P^n = 0$ for $n > a$.*
(2) *Given $K^\bullet$ with $H^n(K^\bullet) = 0$ for $n > a$ there exists a quasi-isomorphism $P^\bullet \to K^\bullet$ with $P^n \in \mathcal{P}$ for all n and $P^n = 0$ for $n > a$.*

Proof. Proof of part (1). Consider the following induction hypothesis IH_n: There are $P^j \in \mathcal{P}$, $j \geq n$, with $P^j = 0$ for $j > a$, maps $d^j : P^j \to P^{j+1}$ for $j \geq n$, and surjective maps $\alpha^j : P^j \to K^j$ for $j \geq n$ such that the diagram

$$\begin{array}{ccccccccc} & & & & P^n & \longrightarrow & P^{n+1} & \longrightarrow & P^{n+2} & \longrightarrow & \ldots \\ & & & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ \ldots & \longrightarrow & K^{n-1} & \longrightarrow & K^n & \longrightarrow & K^{n+1} & \longrightarrow & K^{n+2} & \longrightarrow & \ldots \end{array}$$

is commutative, such that $d^{j+1} \circ d^j = 0$ for $j \geq n$, such that α induces isomorphisms $\mathrm{Ker}(d^j)/\mathrm{Im}(d^{j-1}) \to H^j(K^\bullet)$ for $j > n$, and such that $\alpha : \mathrm{Ker}(d^n) \to \mathrm{Ker}(d^n_K)$ is surjective. Then we choose a surjection

$$P^{n-1} \longrightarrow K^{n-1} \times_{K^n} \mathrm{Ker}(d^n) = K^{n-1} \times_{\mathrm{Ker}(d^n_K)} \mathrm{Ker}(d^n)$$

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with P^{n-1} in $\mathcal{P}$. This allows us to extend the diagram above to

$$\begin{array}{ccccccccc} & & P^{n-1} & \longrightarrow & P^n & \longrightarrow & P^{n+1} & \longrightarrow & P^{n+2} & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ \ldots & \longrightarrow & K^{n-1} & \longrightarrow & K^n & \longrightarrow & K^{n+1} & \longrightarrow & K^{n+2} & \longrightarrow & \ldots \end{array}$$

The reader easily checks that IH_{n-1} holds with this choice.

We finish the proof of (1) as follows. First we note that IH_n is true for $n = a + 1$ since we can just take $P^j = 0$ for $j > a$. Hence we see that proceeding by descending induction we produce a complex $P^\bullet$ with $P^n = 0$ for $n > a$ consisting of objects from $\mathcal{P}$, and a termwise surjective quasi-isomorphism $\alpha : P^\bullet \to K^\bullet$ as desired.

Proof of part (2). The assumption implies that the morphism $\tau_{\leq a}K^\bullet \to K^\bullet$ (Homology, Section 0118) is a quasi-isomorphism. Apply part (1) to find $P^\bullet \to \tau_{\leq a}K^\bullet$. The composition $P^\bullet \to K^\bullet$ is the desired quasi-isomorphism. $\square$